Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Led a production **multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that autonomously processes **financial workflows**, handling **millions of daily transactions** at **99%+ uptime**.
- Built agent workflows across **GPT, Claude, Llama, and Gemini** with **CrewAI** and **AutoGen** alongside LangGraph, covering **tool calling**, memory, and multi-agent planning.
- Built **evaluation harnesses** measuring LLM output quality, safety, **hallucination rates**, and grounding across agent versions, gating releases with **regression suites**.
- Shipped **SSE-based streaming** agent delivery with **guardrails** and real-time **observability**, cutting redundant API calls **90%**.

QureAI | *AI Engineer* Mar 2026 – Present

- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.
- Implemented **GraphRAG** on a **Neo4j** knowledge graph for a regulated healthcare product, and agent workflows on **AWS Bedrock**.
- Built a **voice agent** for the agentic product surface, plus agent workflows on **AWS Bedrock** Agents, Knowledge Bases, and Guardrails.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime**.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.
- Cut production escalations **60%** in one quarter by standardizing API contracts and failure handling, driving **root cause analysis** and post-incident reviews.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.

Projects

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

LangFetch `tinyurl.com/LangFetch`

- **LangGraph** agent that decomposes natural-language **financial queries** into dependency-ordered **SQL** subqueries with **Pinecone** retrieval, served via **FastAPI** as an **MCP** server.

Distributed KV Store `github.com/ganeshhgupta/distributed-kv-store`

- **Raft**-consensus key-value store with **leader election**, **log replication**, **log compaction**, and linearizable reads for fault-tolerant distributed coordination.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Java, Python, SQL, TypeScript

Streaming and Distributed Systems: Apache Kafka, Event-Driven Systems, SQS, Raft Consensus, Log Replication, Log Compaction, Fault Tolerance

Performance and Operations: Throughput and Latency Optimization, Observability, Root-Cause Analysis, Fleet Reliability, Open Source

AWS and Infra: EC2, S3, Lambda, ECS, DynamoDB, Docker, Kubernetes, GitHub Actions CI/CD

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer